The Impact of Loyalty Programs and Coupon Usage on Cart Abandonment: An Experimental Analysis

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1. Research Question

How do loyalty program membership and coupon usage frequency impact the cart abandonment rate of users?

2. Methodology

To answer this research question, this study employs an **Experimental Design and A/B Testing** approach. Specifically, a **Two-Way ANOVA** (Analysis of Variance) is utilized to evaluate the main effects and the interaction effect of loyalty program membership and coupon usage on the cart abandonment rate.

This quantitative method is chosen because it allows for the examination of causal inference between categorical independent variables (loyalty status, coupon usage) and a continuous dependent variable (abandonment rate). The underlying mathematical model for this analysis can be expressed as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 \times X_2) + \epsilon$$

(Where: Y is cart_abandonment_rate, X_1 is loyalty_program_member, X_2 is coupon_usage_frequency, and ϵ is the random error term).

Data Import and Preparation

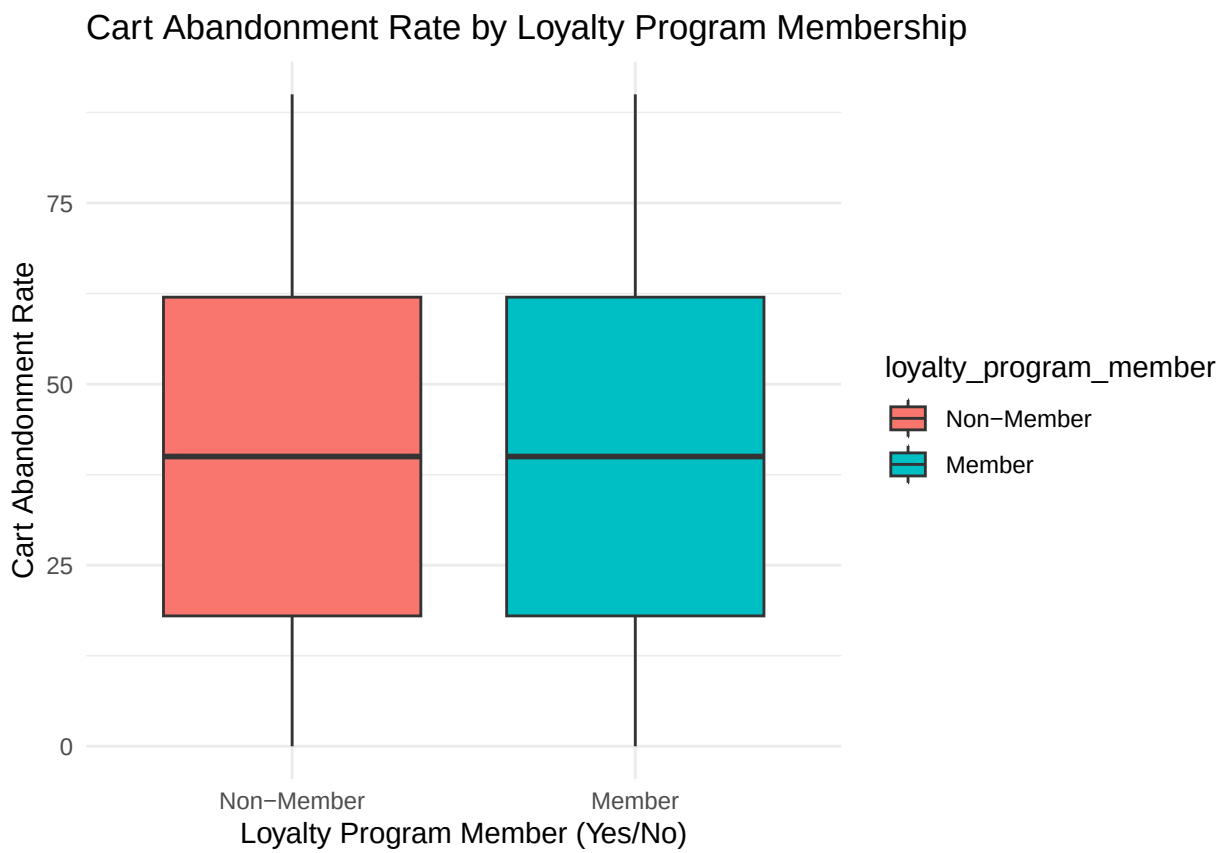
```
# ---------------------------------------------------------
# CLEAN CODE: Step 1 - Data Import and Preprocessing
# ---------------------------------------------------------

# Load required libraries for visualization and data manipulation
library(ggplot2)
library(dplyr)

# 1. Import the E-commerce dataset
ecommerce_data <- read.csv("/Users/nguyenchinghia/Downloads/e_commerce_shopper_behaviour_and_lifestyle.c

# 2. Select the three key variables for the ANOVA model
analysis_data <- ecommerce_data %>%
  select(loyalty_program_member, coupon_usage_frequency, cart_abandonment_rate) %>%
  na.omit() # Remove any missing values to ensure accurate analysis

# 3. Convert grouping variables to Factors and add clear labels
analysis_data$loyalty_program_member <- factor(analysis_data$loyalty_program_member,
```

```r
                                                levels = c(0, 1),
                                                labels = c("Non-Member", "Member"))

analysis_data$coupon_usage_frequency <- as.factor(analysis_data$coupon_usage_frequency)

# Display the first 6 rows of the cleaned dataset
head(analysis_data)
```

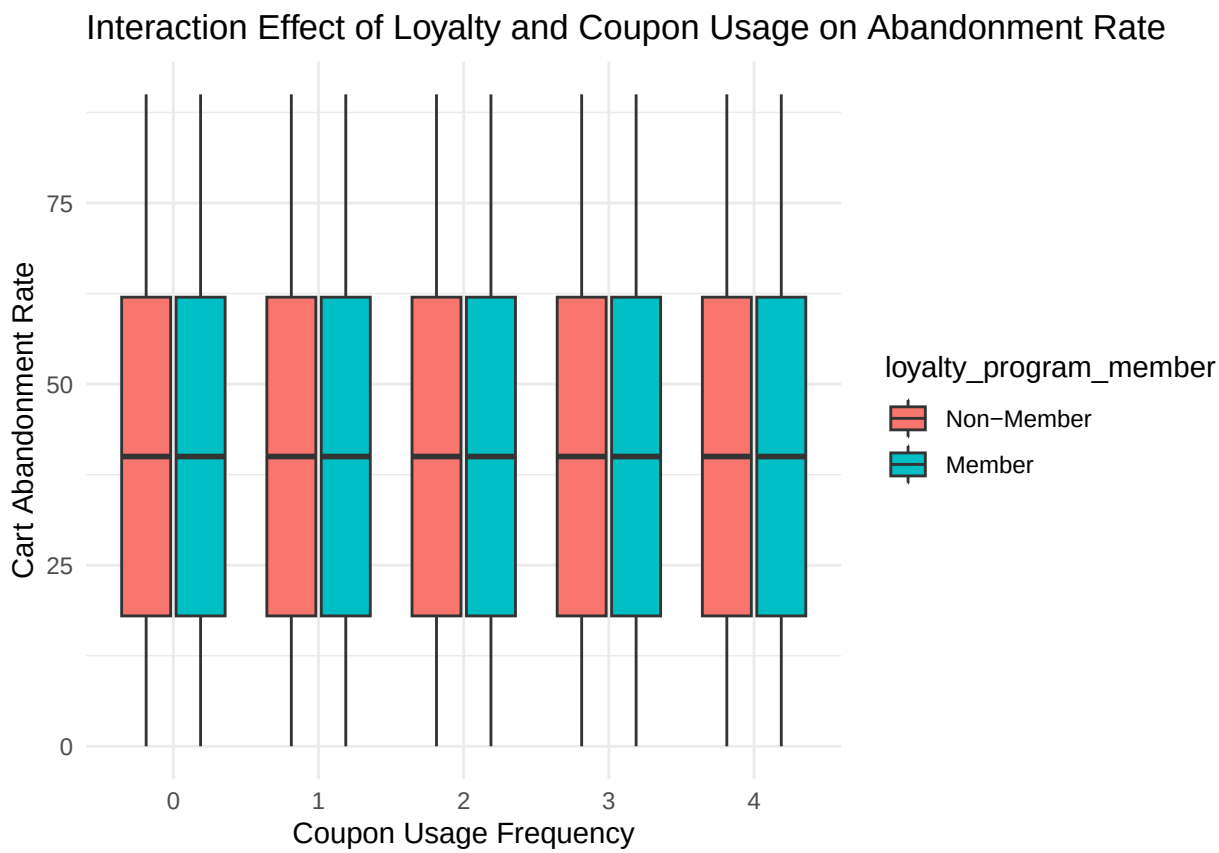
```
##   loyalty_program_member coupon_usage_frequency cart_abandonment_rate
## 1                 Member                      3                     0
## 2                 Member                      0                    28
## 3             Non-Member                      0                    14
## 4                 Member                      2                    11
## 5                 Member                      0                    57
## 6             Non-Member                      3                    46
```

Data Visualization

```r
# ----------------------------------------------------------
# CLEAN CODE: Step 2 - Visualizing Experimental Data
# ----------------------------------------------------------

# Create a Boxplot to visually compare Cart Abandonment Rate between Loyalty Members and Non-members
ggplot(analysis_data, aes(x = loyalty_program_member, y = cart_abandonment_rate, fill = loyalty_program_
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Cart Abandonment Rate by Loyalty Program Membership",
       x = "Loyalty Program Member (Yes/No)",
       y = "Cart Abandonment Rate")
```




```r
# Create an Interaction Plot to examine the combined effect of Loyalty and Coupon Usage
ggplot(analysis_data, aes(x = coupon_usage_frequency, y = cart_abandonment_rate, fill = loyalty_program_
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Interaction Effect of Loyalty and Coupon Usage on Abandonment Rate",
       x = "Coupon Usage Frequency",
       y = "Cart Abandonment Rate")
```



Statistical Analysis: Two-Way ANOVA

```
# ---------------------------------------------------------
# CLEAN CODE: Step 3 - Running the Statistical Model
# ---------------------------------------------------------

# Build a Two-Way ANOVA model to assess main effects and interaction effects (using the * operator)
anova_model <- aov(cart_abandonment_rate ~ loyalty_program_member * coupon_usage_frequency, data = analy

# Display the summary of the results (P-values, F-values) for causal inference
summary(anova_model)
```

```
##                                                Df    Sum Sq Mean Sq F value
## loyalty_program_member                       1e+00        55    54.9   0.085
## coupon_usage_frequency                       4e+00      5952  1487.9   2.300
## loyalty_program_member:coupon_usage_frequency 4e+00      2539   634.7   0.981
## Residuals                                    1e+06 646845724   646.9
##                                              Pr(>F)
## loyalty_program_member                       0.7707
## coupon_usage_frequency                       0.0563 .
## loyalty_program_member:coupon_usage_frequency 0.4163
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3. Findings & Implications

1. Findings:

- **Main Effect 1:** Loyalty program membership does **not** significantly reduce cart abandonment rate (p-value = 0.7707 > 0.05). Both members and non-members abandon their carts at nearly identical rates.
- **Main Effect 2:** Coupon usage frequency shows a marginal, but statistically non-significant impact on cart abandonment rate (p-value = 0.0563 > 0.05).
- **Interaction Effect:** There is **no significant interaction** between being a loyalty member and coupon usage frequency (p-value = 0.4163 > 0.05). The combined promotional efforts do not yield a synergistic effect in retaining the carts.

2. Implications:

Acorrding to the causal inference drawn from this experimental design, **E-commerce platforms should not heavily rely on the current loyalty programs or coupon distributions as the primary strategy to reduce cart abandonment**. In fact, the statistical evidence reveals these promotional tactics are ineffective for this specific problem. Therefore, management should instead redirect their analytical focus and resources toward other potential causes of cart abandonment.